

The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

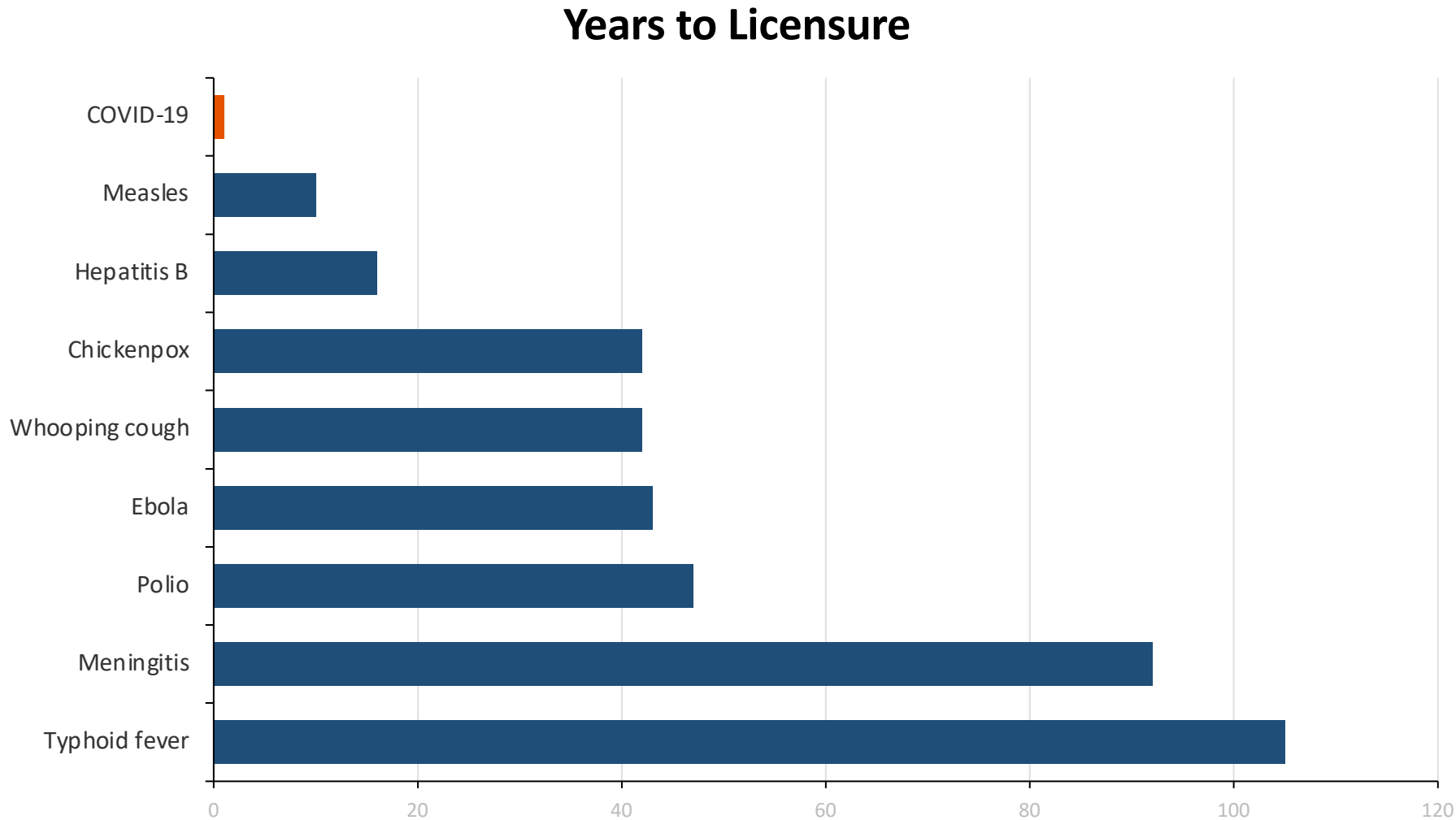
A data-driven overview spanning 1796 – 2020

Source: Our World in Data · Roush & Murphy (2007)

For public health professionals and policy analysts

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure



Modern tools — advanced microscopy, genome sequencing, mRNA technology, and virus-like particles — have enabled faster and more precise vaccine design.

COVID-19: <1 year

mRNA technology & genomic tools collapsed a century-long timeline

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Pre-vaccine vs. post-vaccine annual rates per million population

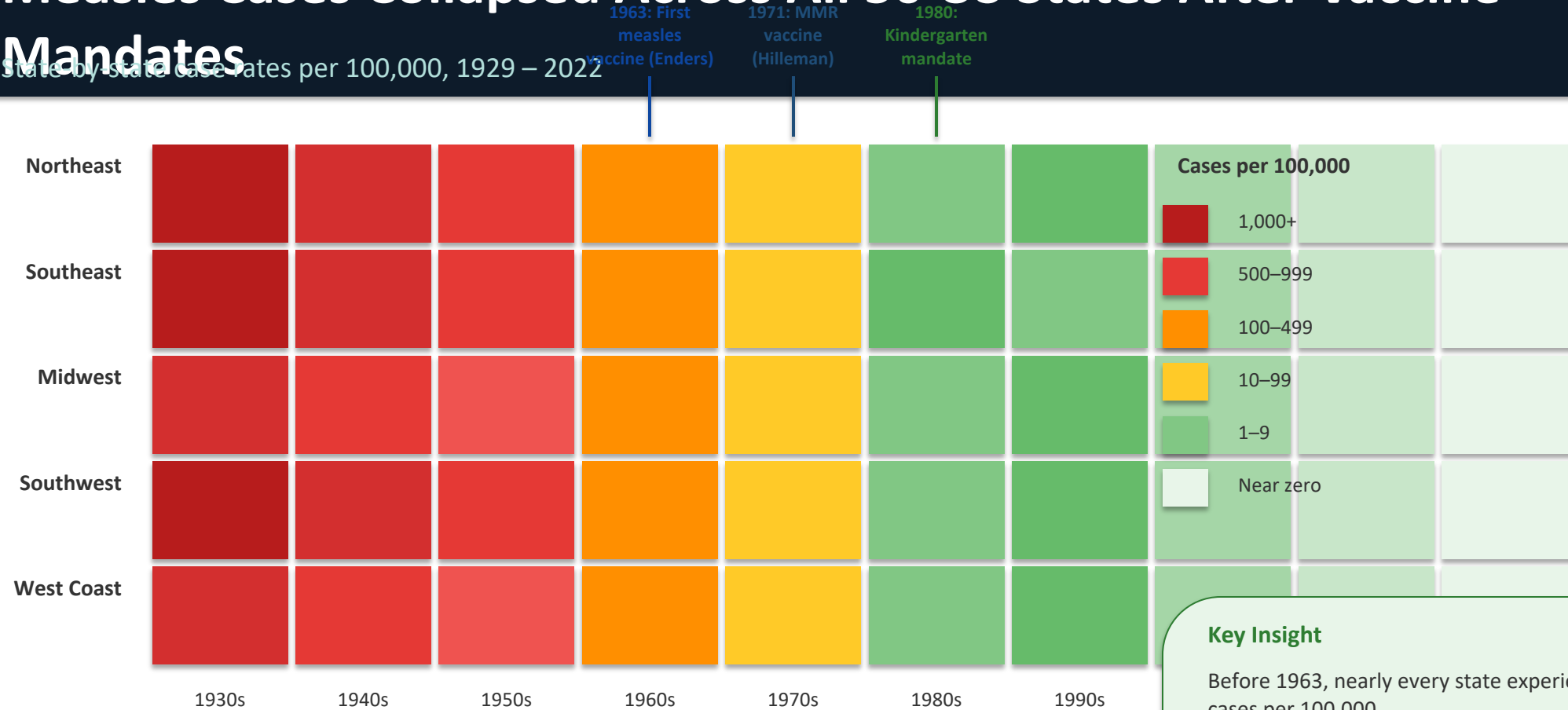
Disease	Pre-Vaccine Cases/million/yr	Post-Vaccine Cases/million/yr	Case Reduction	Pre-Vaccine Deaths/million/yr	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

 100% elimination achieved

 Below 90% — gaps persist

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

State by state case rates per 100,000, 1929 – 2022



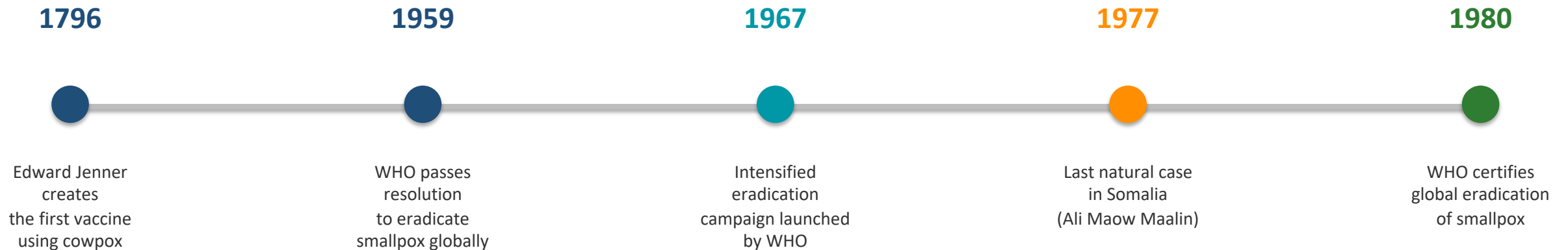
Key Insight

Before 1963, nearly every state experienced 100–1,000+ cases per 100,000.

After the 1980 mandate, cases dropped to near zero across all states, with only sporadic outbreaks.

Smallpox: The Only Human Disease Eradicated Through Vaccination

A global campaign from WHO resolution to certified eradication



Why smallpox was uniquely suited for eradication:

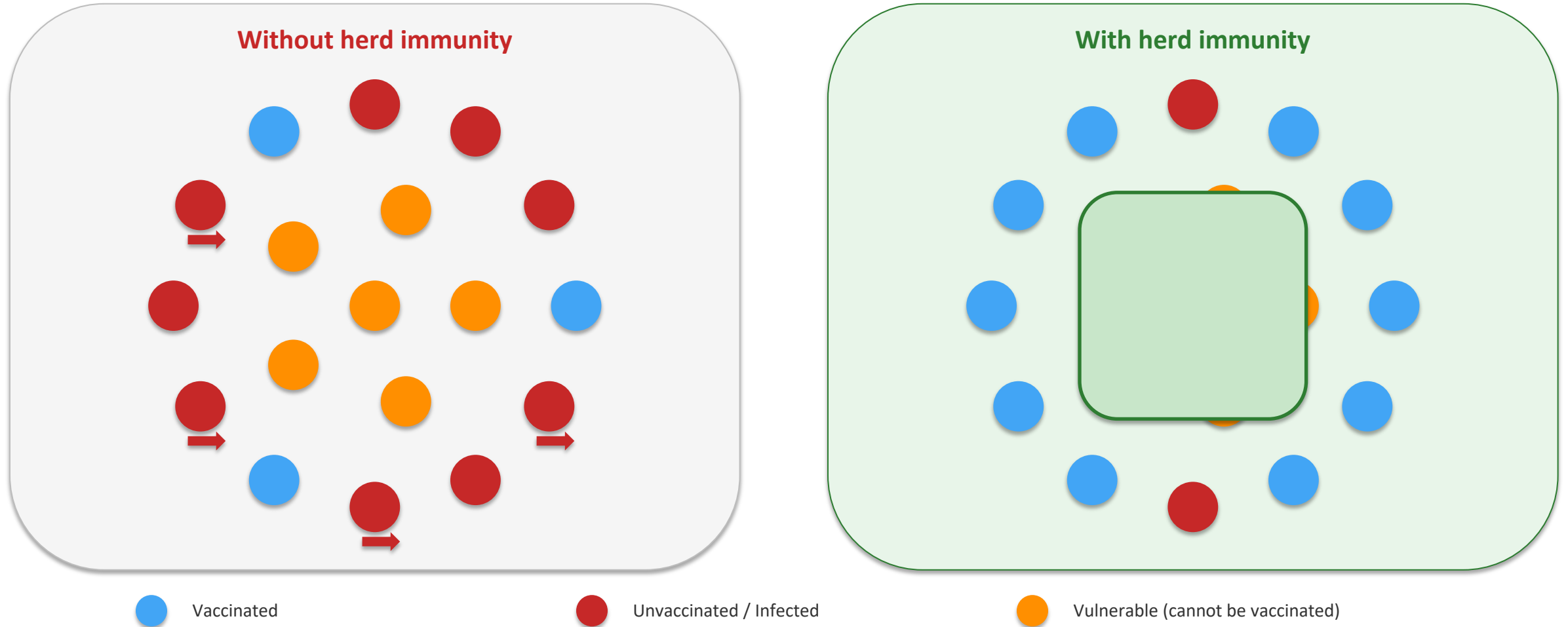
- The virus infected only humans (no animal reservoir)
- Symptoms were easily distinguishable
- The vaccine was cheap and effective

Impact:

- 250 pre-vaccine cases per million/yr in the US → 0
- 100% case reduction and 100% death reduction
- The only human disease ever fully eradicated

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, pathogens cannot spread effectively



Vaccines protect not only individuals but entire communities — shielding newborns, immunocompromised individuals, and cancer patients on chemotherapy.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Diseases where vaccine impact remains incomplete

Pneumococcal Disease

40.5%

case reduction

31.3%

death reduction

233 → 139 cases/million/yr

Hepatitis A

89%

case reduction

88.7%

death reduction

465 → 51 cases/million/yr

Acute Hepatitis B

83.9%

case reduction

83.6%

death reduction

273 → 44 cases/million/yr

Barriers to Full Vaccine Impact

- Access gaps: Not everyone globally has access to vaccines
- Supply shortages: Put children at risk in underserved regions
- Trust challenges: Poor scientific understanding can reduce trust and uptake
- Evolving pathogens: Some diseases require updated or newer vaccine approaches

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

1

100% elimination achieved

Vaccines eliminated cases and deaths for diphtheria, smallpox, and both forms of polio in the US.

2

Development timelines collapsed

From over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.

3

Measles near-eliminated nationwide

Cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate.

4

Smallpox: sole eradication success

Certified in 1980 after a global campaign — the only human disease eradicated through vaccination.

5

Coverage gaps persist

Call to Action: Improve access, expand coverage, and build trust in vaccination to close remaining gaps.

Global health systems must invest in equitable distribution, supply chain resilience, and science communication.